# NADEL VANISHING ON SINGULAR COMPLEX SPACES VIA THE GRAUERT–RIEMENSCHNEIDER CANONICAL SHEAF

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Abstract. We formulate Nadel vanishing for reduced, pure-dimensional, weakly pseudoconvex Kähler complex spaces. The correct replacement for $K_X \otimes \mathcal{I}(h)$ is the weighted Grauert–Riemenschneider canonical sheaf $\omega_X^{\mathrm{GR}}(h)$. We record its description on a resolution and derive the vanishing theorem from two stated analytic inputs: a weighted $L^2$-Dolbeault resolution and a global $\bar\partial$ estimate. The proof makes explicit the positivity required near the singular locus. We also explain why pullback to a resolution does not by itself prove vanishing and give a cusp example showing the failure of a naive tensor-product formulation.

## 1. Relevant results and scope

Demailly's Remark 5.17 in [Dem00] introduces the integrable canonical module on a singular space, records its transformation under modifications, and formulates the corresponding Nadel vanishing theorem. It also stresses that the canonical factor and the multiplier condition must be treated together. Li [Li21, Remark 2.2 and Theorem 2.3] uses the notation $\omega_X^{\mathrm{GR}}(\varphi)$ and gives an explicit singular-space formulation.

For the analytic proof, Shentu–Zhao [SZ20, Theorem 1.3] provide a weighted $L^2$-Dolbeault resolution of a direct image from a resolution of singularities. Watanabe [Wat26, Theorems 1.1, 1.3, and 1.4] gives a weighted fine resolution, a global $L^2$ estimate, and Nadel vanishing, respectively. The global estimate used below is taken from the cited July 15, 2026 version of that preprint; it is stated as an analytic input, not reproved here. Finally, Matsumura [Mat18, Corollary 1.5] supplies relative Grauert–Riemenschneider vanishing with multiplier ideals, which explains the resolution-theoretic cohomology comparison in Section 6.

Throughout, complex spaces are reduced and paracompact, and $X$ has pure complex dimension $n$. We write $X_{\mathrm{reg}}$ and $X_{\mathrm{sing}}$ for its regular and singular loci. A smooth function on $X$ is understood locally through an embedding into a complex manifold. A Kähler metric on $X$ is understood in the usual complex-space sense; on $X_{\mathrm{reg}}$ it determines a Kähler form $\omega$, locally comparable to an ambient Hermitian metric.

We call $X$ *weakly pseudoconvex* if it admits a smooth plurisubharmonic exhaustion $\Psi : X \to [0,\infty)$, meaning that $\{\Psi < c\} \Subset X$ for every $c$. Compact spaces are included by taking $\Psi$ constant. No normality, $\mathbb{Q}$-Gorenstein, or rational-singularity hypothesis is imposed. The Kähler, exhaustion, and curvature hypotheses, however, are part of the vanishing theorem and are not omitted.

## 2. The weighted canonical sheaf

**Definition 2.1.** Let $\varphi \in \mathrm{Psh}(X)$, with $\varphi$ not identically $-\infty$ on any irreducible component. The *weighted Grauert–Riemenschneider canonical sheaf* is defined by

$$\omega_X^{\mathrm{GR}}(\varphi)(U) = \Big\{ \sigma \in H^0(U \cap X_{\mathrm{reg}}, K_{X_{\mathrm{reg}}}) : \int_{V \cap X_{\mathrm{reg}}} \sqrt{-1}^{\,n^2} \sigma \wedge \overline{\sigma}\, e^{-2\varphi} < \infty \quad \text{for every } V \Subset U \Big\}. \tag{2.1}$$

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The integrability condition is imposed up to $X_{\mathrm{sing}}$, not just on compact subsets of $X_{\mathrm{reg}}$. In particular, $\omega_X^{\mathrm{GR}}(0) = \omega_X^{\mathrm{GR}}$, the usual Grauert–Riemenschneider canonical sheaf.

This is the construction in [Dem00, Remark 5.17] and [Li21, Remark 2.2]. Its use of top-degree forms makes the integrability condition intrinsic: it does not depend on a choice of background volume form.

Let $L$ be a holomorphic line bundle on $X$, identified with its sheaf of holomorphic sections. A singular Hermitian metric $h$ is written in a local holomorphic frame $e$ as

$$|e|_h^2 = e^{-2\varphi}, \qquad \sqrt{-1}\Theta_{L,h} = 2\sqrt{-1}\partial\bar{\partial}\varphi. \tag{2.2}$$

Unless stated otherwise, its local weights are plurisubharmonic on $X$ and not identically $-\infty$ on any local irreducible component. If $e' = ge$ is another frame, then $\varphi' = \varphi - \log|g|$. Since $g$ is a holomorphic unit, the two weights differ by a locally bounded function and give the same integrability sheaf. We may therefore write $\omega_X^{\mathrm{GR}}(h)$ for the sheaf locally defined by $\omega_X^{\mathrm{GR}}(\varphi)$, and set

$$\mathcal{F}_{L,h} := \omega_X^{\mathrm{GR}}(h) \otimes_{\mathcal{O}_X} L. \tag{2.3}$$

Equivalently, $\mathcal{F}_{L,h}$ consists of holomorphic $L$-valued top forms on $X_{\mathrm{reg}}$ that are locally square-integrable with respect to $h$, including near singular points.

**Proposition 2.2** (Transformation rule)**.** *Let $\pi : \widetilde{X} \to X$ be a resolution of singularities that is an isomorphism over $X_{\mathrm{reg}}$. Then*

$$\omega_X^{\mathrm{GR}}(\varphi) \simeq \pi_*\big(K_{\widetilde{X}} \otimes \mathcal{I}(\varphi \circ \pi)\big), \tag{2.4}$$

$$\mathcal{F}_{L,h} \simeq \pi_*\big(K_{\widetilde{X}} \otimes \pi^*L \otimes \mathcal{I}(\pi^*h)\big). \tag{2.5}$$

*Here $\mathcal{I}$ is the ordinary multiplier ideal on the smooth space $\widetilde{X}$, using the convention $e^{-2\varphi}$. In particular, these weighted canonical sheaves are coherent.*

*Proof.* The transformation rule is recorded in [Li21, Remark 2.2]; we recall its local mechanism. Away from $\pi^{-1}(X_{\mathrm{sing}})$, pullback identifies the integrals of weighted top forms by change of variables. A form satisfying (2.1) pulls back to a holomorphic top form on the complement of that analytic set. Since $\varphi \circ \pi$ is locally bounded above, weighted square-integrability implies unweighted square-integrability. The $L^2$ removable-singularity theorem for holomorphic functions, applied to the coefficients of the top form, extends it holomorphically across the analytic set. The weighted condition then says exactly that its coefficients belong to $\mathcal{I}(\varphi \circ \pi)$.

Conversely, such an upstairs form restricts and descends over $X_{\mathrm{reg}}$. Properness and change of variables give the required local integrability downstairs. This proves (2.4); the same argument for local frames of $L$, or the projection formula, proves (2.5). Coherence follows from coherence of smooth multiplier ideals and the proper direct image theorem. □

## 3. Statement of the vanishing theorem

**Theorem 3.1** (Nadel vanishing with the weighted GR sheaf)**.** *Let $(X, \omega)$ be a reduced, pure $n$-dimensional, weakly pseudoconvex Kähler complex space. Let $L$ carry a singular Hermitian metric $h$ with plurisubharmonic local weights on $X$. Suppose there is a positive continuous function $\varepsilon : X \to (0, \infty)$ such that*

$$\sqrt{-1}\Theta_{L,h} \geq \varepsilon\omega \quad \text{on } X_{\mathrm{reg}} \tag{3.1}$$

*in the sense of currents. Then*

$$\boxed{H^q\big(X, \omega_X^{\mathrm{GR}}(h) \otimes L\big) = 0 \quad \text{for every } q \geq 1.} \tag{3.2}$$

This formulation follows from the analytic inputs in Section 4; compare [Wat26, Theorem 1.4]. A standard sufficient positivity condition is that the local weights be strictly plurisubharmonic on the complex space, in the local ambient sense. Such weights give a lower bound as in (3.1) with a positive smooth function; see [Wat26, Proposition 3.1].

*Remark* 3.2 (Positivity at the singular locus). The function $\varepsilon$ is positive on *all of* $X$. Thus it has a positive lower bound on every compact subset, including those meeting $X_{\mathrm{sing}}$. Li's formulation [Li21, Theorem 2.3] states a positive continuous bound on $X_{\mathrm{reg}}$. The theorem proved here deliberately uses the stronger hypothesis above: positivity only on $X_{\mathrm{reg}}$, with the bound possibly tending to zero at singular points, does not justify the compact-set integrability step in our proof. Compare the observation following [Wat26, Theorem 6.4].

## 4. Two analytic inputs

For $0 \le q \le n$, let $\mathscr{A}^{n,q}_{L,h}$ be the sheaf whose sections on $U \subset X$ are measurable $L$-valued $(n,q)$-forms $u$ on $U \cap X_{\mathrm{reg}}$ such that both $u$ and $\bar\partial u$ are locally $L^2$ with respect to $(h,\omega)$ up to $X_{\mathrm{sing}}$. The derivative is the maximal distributional $\bar\partial$ on $X_{\mathrm{reg}}$. Thus local integrability means finiteness on $V \cap X_{\mathrm{reg}}$ for every $V \Subset U$.

**Proposition 4.1** (Analytic input A: weighted Dolbeault resolution)**.** *Under the local hypotheses above, the sequence*

$$0 \longrightarrow \mathcal{F}_{L,h} \longrightarrow \mathscr{A}^{n,0}_{L,h} \xrightarrow{\bar\partial} \mathscr{A}^{n,1}_{L,h} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathscr{A}^{n,n}_{L,h} \longrightarrow 0 \tag{4.1}$$

*is an exact complex of fine sheaves after its initial term. Consequently,*

$$H^q(X, \mathcal{F}_{L,h}) \simeq H^q\big(\Gamma(X, \mathscr{A}^{n,\bullet}_{L,h}), \bar\partial\big). \tag{4.2}$$

This is the specialization of [SZ20, Theorem 1.3] to the pullback bundle $(\pi^*L, \pi^*h)$; it is also covered by [Wat26, Theorem 1.1]. Indeed, near any point choose a relatively compact local embedding into $\mathbb{C}^N$. A sufficiently large multiple of $|z|^2$ is a bounded strictly plurisubharmonic local potential dominating the given metric, while $\pi^*h$ has semipositive curvature. These verify the local metric hypotheses of the cited theorem. The initial sheaf is identified by Proposition 2.2. Smooth partitions of unity preserve the maximal $\bar\partial$ domain, so the remaining sheaves are fine.

**Proposition 4.2** (Analytic input B: global estimate)**.** *In the setting of Theorem 3.1, assume $\varepsilon$ is smooth. Let $1 \le q \le n$ and let $f$ be a locally $L^2$, $L$-valued $(n,q)$-form on $X$, with $\bar\partial f = 0$ on $X_{\mathrm{reg}}$. If*

$$\int_{X_{\mathrm{reg}}} \varepsilon^{-1} |f|^2_{h,\omega}\, dV_\omega < \infty, \tag{4.3}$$

*then there is an $L$-valued $(n,q-1)$-form $u$ satisfying $\bar\partial u = f$ on $X_{\mathrm{reg}}$ and*

$$\int_{X_{\mathrm{reg}}} |u|^2_{h,\omega}\, dV_\omega \le \frac{1}{q} \int_{X_{\mathrm{reg}}} \varepsilon^{-1} |f|^2_{h,\omega}\, dV_\omega. \tag{4.4}$$

This is the line-bundle case of [Wat26, Theorem 1.3]. All integrals above are taken on the regular locus, with integrability required near the singular locus as well. We use this result directly; we do not assume that the given metric on $X_{\mathrm{reg}}$ is complete.

## 5. Proof of the vanishing theorem

The role of an exhaustion is to turn a locally $L^2$ cohomology representative into a globally integrable one without altering its local integrability sheaf.

**Lemma 5.1** (An integrable exhaustion weight)**.** *Let $\Psi : X \to [0,\infty)$ be a continuous exhaustion and let $\mu$ be a positive Borel measure finite on compact subsets of $X$. There exists a smooth increasing convex function $\chi : \mathbb{R} \to \mathbb{R}$ such that*

$$\int_X e^{-\chi(\Psi)}\, d\mu < \infty. \tag{5.1}$$

*Proof.* Set $a_j = \mu(\{j \le \Psi < j+1\}) < \infty$ for $j \ge 0$. Choose a smooth increasing convex function satisfying

$$\chi(j) \ge \log(1 + a_j) + j \log 2 \qquad (j \ge 0).$$

Such a function is obtained by constructing an increasing convex polygonal majorant and smoothing it from above. On the $j$th shell, $\chi(\Psi) \ge \chi(j)$, so

$$\int_X e^{-\chi(\Psi)}\, d\mu \le \sum_{j=0}^{\infty} a_j e^{-\chi(j)} \le \sum_{j=0}^{\infty} 2^{-j} < \infty. \qquad \square$$

*Proof of Theorem 3.1.* A positive continuous function on $X$ has a positive smooth minorant, constructed with a locally finite smooth partition of unity. Replacing $\varepsilon$ by such a minorant, we assume it is smooth. For $q > n$ the conclusion follows already from the length of the fine resolution (4.1). Fix $1 \le q \le n$.

By (4.2), a class in $H^q(X, \mathcal{F}_{L,h})$ has a representative

$$f \in \Gamma(X, \mathscr{A}^{n,q}_{L,h}), \qquad \bar\partial f = 0.$$

Consider the measure, extended by zero on $X_{\mathrm{sing}}$,

$$d\mu = \varepsilon^{-1} |f|^2_{h,\omega}\, dV_\omega.$$

It is finite on compact subsets: $f$ is locally $L^2$ up to the singular locus and $\varepsilon$ has a positive minimum on each compact set. Choose a smooth plurisubharmonic exhaustion $\Psi$. By Lemma 5.1, there is a smooth increasing convex $\chi$ with

$$\int_{X_{\mathrm{reg}}} \varepsilon^{-1} |f|^2_{h,\omega} e^{-\chi(\Psi)}\, dV_\omega < \infty. \tag{5.2}$$

Define a new metric on the same line bundle by

$$h_\chi = h\, e^{-\chi(\Psi)}.$$

Its local weights are $\varphi + \frac{1}{2}\chi(\Psi)$, hence remain plurisubharmonic. Moreover,

$$\begin{aligned} \sqrt{-1}\Theta_{L,h_\chi} &= \sqrt{-1}\Theta_{L,h} + \sqrt{-1}\partial\bar\partial\chi(\Psi) \\ &= \sqrt{-1}\Theta_{L,h} + \chi'(\Psi)\sqrt{-1}\partial\bar\partial\Psi + \chi''(\Psi)\sqrt{-1}\partial\Psi \wedge \bar\partial\Psi \\ &\ge \varepsilon\omega. \end{aligned} \tag{5.3}$$

On every compact subset, $e^{-\chi(\Psi)}$ is bounded above and below by positive constants. Consequently,

$$\mathcal{F}_{L,h_\chi} = \mathcal{F}_{L,h}, \qquad \mathscr{A}^{n,\bullet}_{L,h_\chi} = \mathscr{A}^{n,\bullet}_{L,h}. \tag{5.4}$$

Apply Proposition 4.2 to the metric $h_\chi$ and the form $f$. Equations (5.2) and (5.3) yield $u$ with $\bar\partial u = f$ and

$$\int_{X_{\mathrm{reg}}} |u|^2_{h,\omega} e^{-\chi(\Psi)}\, dV_\omega \le \frac{1}{q} \int_{X_{\mathrm{reg}}} \varepsilon^{-1} |f|^2_{h,\omega} e^{-\chi(\Psi)}\, dV_\omega < \infty.$$

Local equivalence of $h_\chi$ and $h$ implies that $u$ is locally $L^2$ for $h$, up to the singular locus. Since $\bar\partial u = f$ is locally $L^2$ as well, $u \in \Gamma(X, \mathscr{A}^{n,q-1}_{L,h})$. Thus $f$ is exact in the complex computing sheaf cohomology, and the original class vanishes. $\square$

## 6. What a resolution argument gives

Choose a resolution $\pi : \widetilde{X} \to X$ that is locally projective over $X$, and put

$$\mathcal{G} = K_{\widetilde{X}} \otimes \pi^* L \otimes \mathcal{I}(\pi^* h).$$

Since the local weights of $h$ are plurisubharmonic on $X$, the metric $\pi^* h$ has semipositive curvature on $\widetilde{X}$. Matsumura's relative vanishing [Mat18, Corollary 1.5], applied locally on the base and componentwise if necessary, gives

$$R^j \pi_* \mathcal{G} = 0 \qquad (j > 0). \tag{6.1}$$

Indeed, that result gives vanishing in degrees greater than the dimension of the general fiber, which is zero here. Together with (2.5), the Leray spectral sequence yields

$$H^q(X, \mathcal{F}_{L,h}) \simeq H^q(\widetilde{X}, \mathcal{G}). \tag{6.2}$$

*Remark* 6.1 (The missing positivity in a naive pullback proof). Equation (6.2) is a cohomology comparison, not yet a vanishing proof. Even if one has the pulled-back inequality

$$\sqrt{-1}\Theta_{\pi^* L, \pi^* h} \geq (\varepsilon \circ \pi)\pi^* \omega,$$

the form $\pi^* \omega$ is only semipositive: it can degenerate where $d\pi$ has a kernel. Thus it is not, in general, a Kähler metric on $\widetilde{X}$, and ordinary smooth Nadel vanishing cannot be invoked from this inequality alone. One needs an additional positivity argument, or the direct singular-space estimate used above; compare [Wat26, Remark 6.5].

## 7. Specializations and the tensor-product issue

**Corollary 7.1** (Smooth spaces and locally bounded metrics)**.** *Under the hypotheses of Theorem 3.1, the following hold. If $X$ is smooth, then*

$$\omega_X^{\mathrm{GR}}(h) = K_X \otimes \mathcal{I}(h), \qquad H^q(X, K_X \otimes L \otimes \mathcal{I}(h)) = 0 \quad (q \geq 1).$$

*If the local weights of $h$ are locally bounded, in particular if $h$ is smooth, then*

$$\omega_X^{\mathrm{GR}}(h) = \omega_X^{\mathrm{GR}}, \qquad H^q(X, \omega_X^{\mathrm{GR}} \otimes L) = 0 \quad (q \geq 1).$$

*For compact $X$, these statements apply with a constant exhaustion.*

*Proof.* On a manifold, writing a top form in local coordinates identifies its weighted integrability with membership of its coefficient in $\mathcal{I}(h)$. For a locally bounded weight, multiplication by $e^{-2\varphi}$ does not change local integrability. The assertions now follow from Theorem 3.1. □

In general, however, $\omega_X^{\mathrm{GR}}(h)$ must not be replaced without justification by $\omega_X^{\mathrm{GR}}$ tensored with a separately defined multiplier ideal on $X$. The following calculation exhibits the problem for the direct volume-based definition.

*Example* 7.2 (The cusp). Work near the origin of the cusp

$$X = \{(x, y) \in \mathbb{C}^2 : y^2 = x^3\}, \qquad \nu(t) = (t^2, t^3),$$

whose normalization is a small disc $\Delta$. Set $\varphi = \frac{1}{2} \log |x|$, so that $\varphi \circ \nu = \log |t|$. For the area measure induced by the Euclidean metric,

$$\nu^* dV_X = (4|t|^2 + 9|t|^4)\, dA(t) \asymp |t|^2 dA(t).$$

Define the volume-based ideal by

$$\mathcal{I}_{\mathrm{vol}}(\varphi) = \{g \in \mathcal{O}_X : |g|^2 e^{-2\varphi} \in L^1_{\mathrm{loc}}(dV_X)\}.$$

The constant function 1 satisfies this condition, since

$$e^{-2\varphi \circ \nu} \nu^* dV_X = (4 + 9|t|^2)\, dA(t).$$

Thus $\mathcal{I}_{\mathrm{vol}}(\varphi) = \mathcal{O}_X$.

For a holomorphic top form upstairs, the weighted condition is instead

$$\int_{|t|<r} |a(t)|^2 |t|^{-2}\, dA(t) < \infty \quad \text{for the form } a(t)\, dt.$$

For sufficiently small $r > 0$, it holds exactly when $a(0) = 0$: if $a$ has vanishing order $k$, the radial integral is comparable to $\int_0^r \rho^{2k-1}\, d\rho$, finite exactly when $k \geq 1$. Consequently,

$$\omega_X^{\mathrm{GR}} = \nu_*(\mathcal{O}_\Delta\, dt), \qquad \omega_X^{\mathrm{GR}}(\varphi) = \nu_*(t\mathcal{O}_\Delta\, dt),$$

and therefore

$$\omega_X^{\mathrm{GR}}(\varphi) \neq \omega_X^{\mathrm{GR}} \otimes \mathcal{I}_{\mathrm{vol}}(\varphi). \tag{7.1}$$

This example concerns the explicitly defined volume-based ideal; it does not identify or conflate the different multiplier-ideal conventions available on singular spaces.

The appropriate singular analogue of the sheaf in Nadel vanishing is therefore the single weighted canonical module

$$K_X \otimes L \otimes \mathcal{I}(h) \quad \rightsquigarrow \quad \omega_X^{\mathrm{GR}}(h) \otimes L \simeq \pi_*\big(K_{\widetilde{X}} \otimes \pi^* L \otimes \mathcal{I}(\pi^* h)\big),$$

with higher-cohomology vanishing under the hypotheses of Theorem 3.1.

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